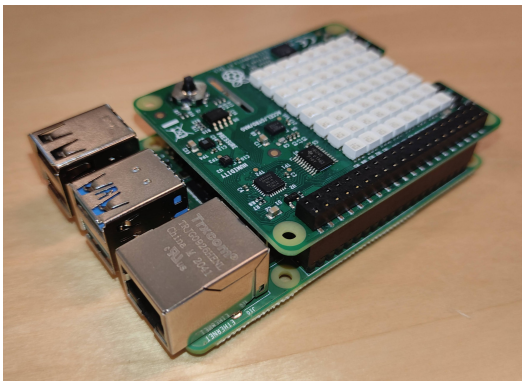
TDT4258 Low-Level Programming
Autumn 2026

Raspberry Pi 4 w/Sense Hat

Setup Guide



Version 1.0

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In Lab 3 you will program a Raspberry Pi 4 (RPi4) with an extension board (Sense Hat). Since we will use the RPi4 to run example programs shown in the lectures, we will in this semester lend out the RPi4s already from the second week of the course. This document describes how to set up the RPi4 to be able to do Lab 3 and run the code examples.

1 Setting up the Raspberry Pi 4

You will be provided with a Raspberry Pi 4, a power supply, some additional cables ,an SD card for the operating system and a USB SD card reader. You have to flash an operating system image onto the SD card and find a suitable way to connect to the Raspberry Pi 4. You will have two main different options to work with the Raspberry Pi 4:

1. You can use it as a normal desktop and connect a screen, keyboard and mouse to it.
2. You can also work over the terminal on the Raspberry Pi 4, for which you need a WiFi connection. Note that eduroam will not work and that you should **NOT connect it to a network socket with a cable in the IT-building, or in other NTNU buildings** since it might trigger a IT security policy that will block that socket. An alternative connection can be to set up your mobile phone as a personal hotspot. Se also further details in Section 1.2.

For the first step, you have to flash the operating system onto the SD card. Do not use whatever is installed on the SD card since it is most probably not the correct OS with the correct configuration.

To flash the image onto the SD card, go to
https://www.raspberrypi.com/software/, download and install the Raspberry Pi Imager on your local machine. Plug in your SD card and open the Raspberry Pi Imager. On Linux you may have to open the program with sudo. This guide is made with version v2.0.11.1.

1.1 Raspberry Pi OS

Once Raspberry Pi Imager is open, set the device to 'Raspberry Pi 4' (see Figure 1).

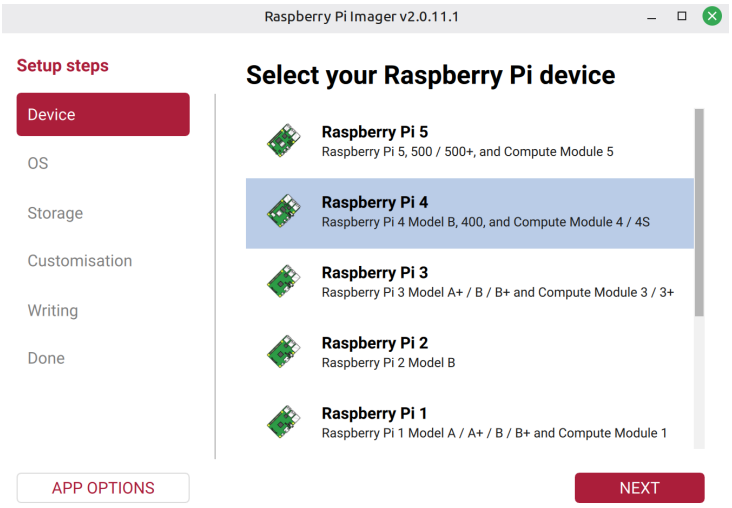


Figure 1: Select Raspberry Pi 4

Press Next and set the Operating System to **'Raspberry Pi OS (Legacy, 32-bit)'** (see Figure 2).

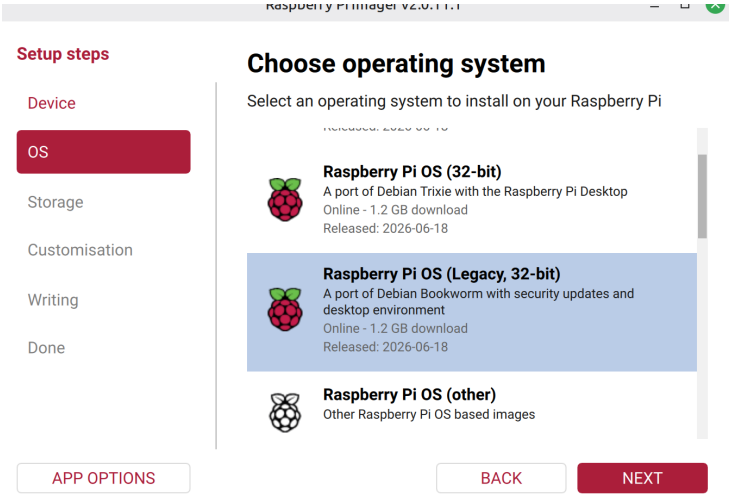


Figure 2: Select Raspberry Pi OS (Legacy, 32-bit)

Press Next to Storage, and select the SD card reader (see Figure 3).

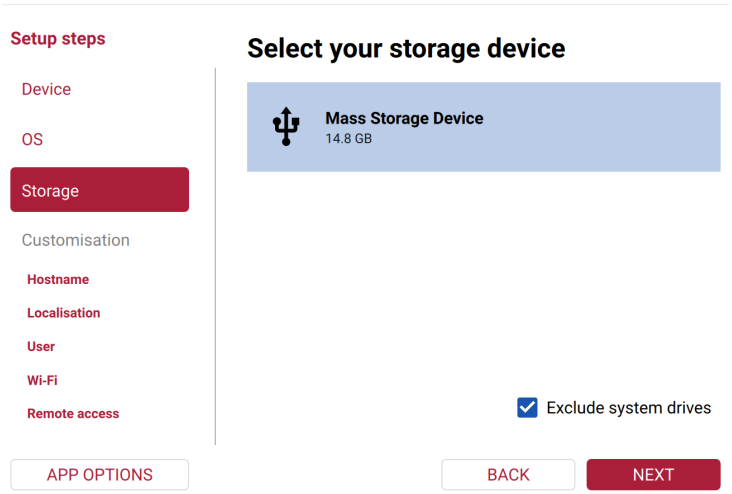


Figure 3: Select Mass Storage Device (your SD card)

Next under Customisation hostname, give your Pi a hostname, this is used to identify your device on the relevant network (see Figure 4).

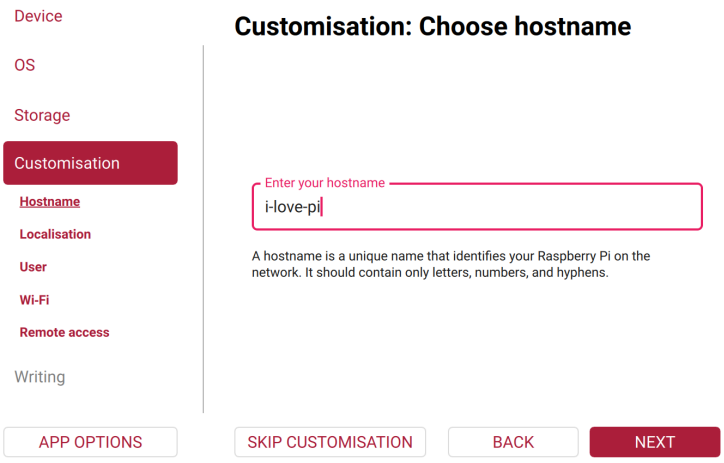


Figure 4: Selection of hostname

Then under user create a username and a password. You can connect to your Pi using SSH and directly login to your user (see Figure 5).

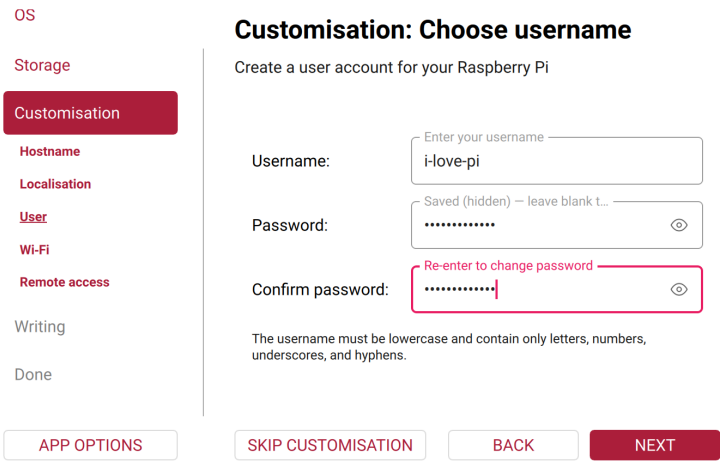


Figure 5: Creation of user

Select the Wi-Fi menu keep the selection to secure network. Under SSID enter the name of your network, (note that eduroam will not work). Then enter the network password and press Next (see Figure 6).

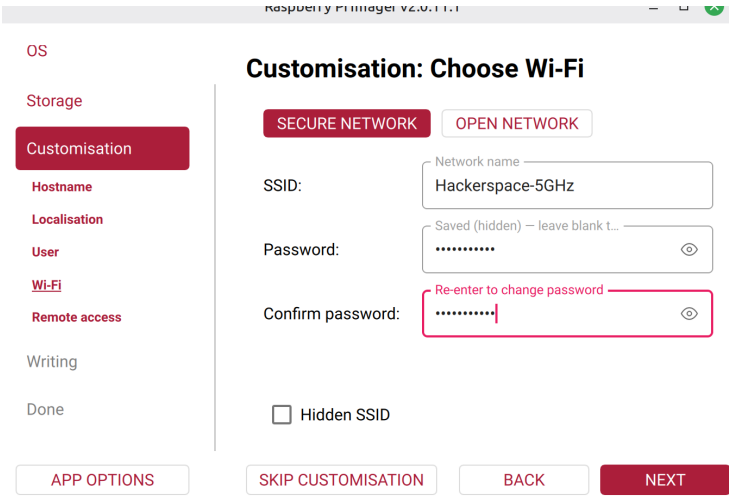


Figure 6: Connecting to Wi-Fi

Press next so that you are in the Remote access menu. Toggle enable SSH. Select "Use password authentication" as it is the easiest. When you connect via SSH, it uses the same password and the user you created earlier (see Figure 7).

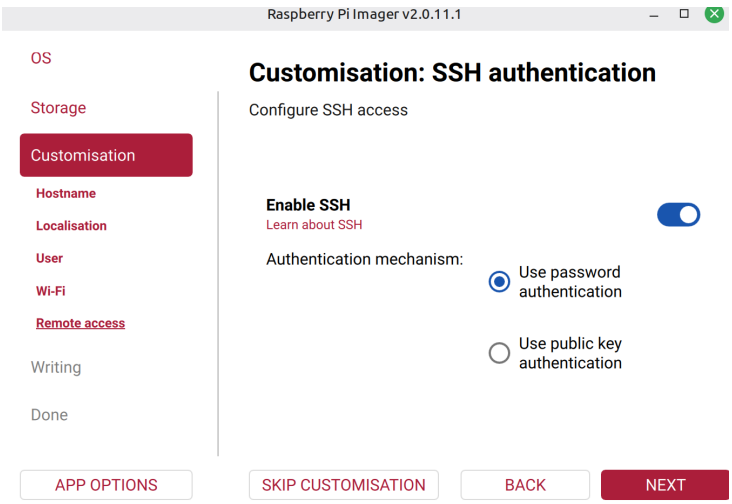


Figure 7: Enabling SSH

When you are done the summary should look something like in figure 8.

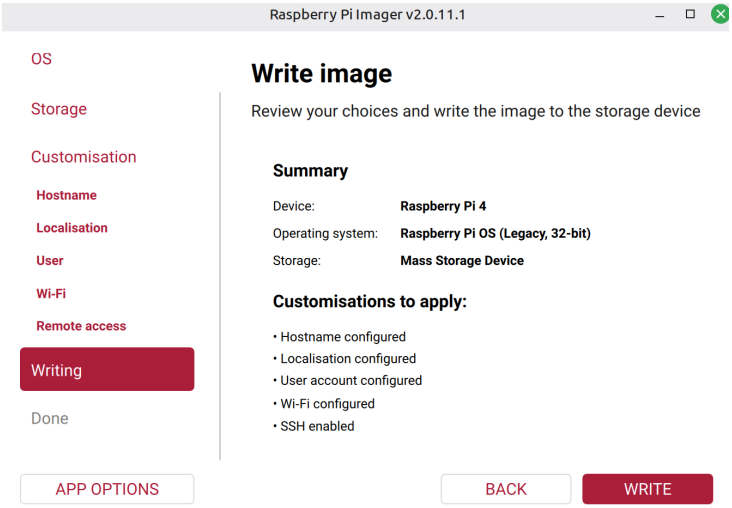


Figure 8: Summary of configurations.

Press the write button and confirm that you want to overwrite the content of the SD card. Unplug the SD card after writing the image has finished and insert it on the back of the Raspberry Pi 4. The side of the SD card with print (eg. 16 GB) should be down (away from the RPi4-unit).

You are provided with a Raspberry Pi Sense Hat extension board, the Sense Hat should already be connected your Raspberry Pi 4. In the case it is not,

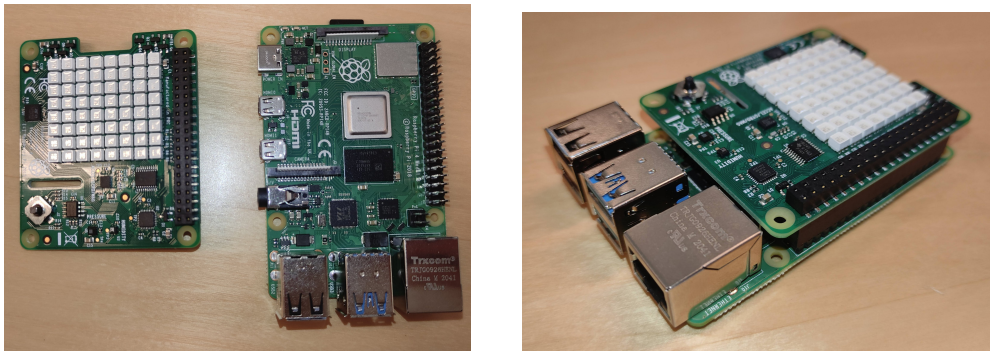


Figure 9: Sense Hat extension board and Raspberry Pi 4, disassembled (left), and assembled (right).

connect it to the GPIO pin header so that the extension board covers the Raspberry Pi 4 PCB like seen in Figure 9.

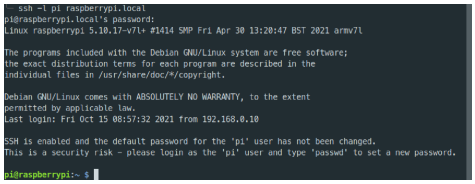
1.2 Working on the Raspberry Pi 4

Connect the power supply to the Raspberry Pi 4 and it will automatically start to boot. The Sense Hat board lights up the LED matrix in rainbow colors. After booting has finished, the Sense Hat LED matrix shows a white square.

If you want to work on the Raspberry Pi from your own computer, you have to connect it to your local network via Ethernet or WiFi. You can find the IP address of the Raspberry Pi in your router or connect to it with the hostname that you have set during the flashing process. You will need an SSH client on your system (preinstalled on Linux and MacOS; for Windows you can use the Linux Subsystem on Windows). If you are working on Windows and none of these options are available to you, your can install PuTTY or any other SSH client that is available for Windows. In Figure 10 and 11 you can see the different methods for connecting a Raspberry Pi. To log in, use the username and password that entered during the SD card setup.



SSH connection via IP



SSH connection via hostname, where the username is pi, and the hostname is raspberrypi

Figure 10: Connecting to the Raspberry Pi.

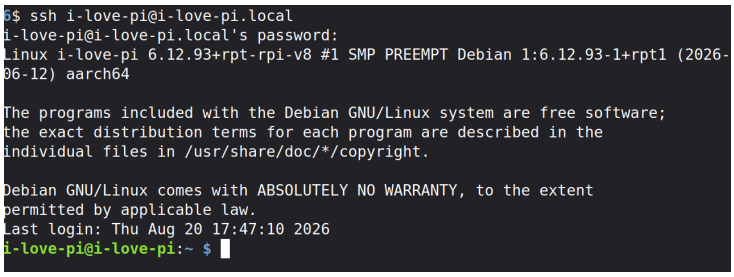


Figure 11: SSH connection to PI using hostname and username, where the username and the hostname is i-love-pi

Further details about how to use the RPi4 and interact with the sense hat are given in the description of Lab 3.